**PRACTICAL SUBMISSION RECORD- A.Y. 2024-25**

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| **Class: SYMCA Div: B**<br>**Semester: III** | **Course Code: MCA01604**<br>**Course Name: Data Science Laboratory** | **Batch: S1** |
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| **CO No: CO605.4** | **Mini Project**: Automated News Classification System | |

## 1. Objective

The primary objective of this project is to design, build, and evaluate a machine learning model capable of automatically classifying news articles into four distinct categories: **World, Sports, Business, and Sci/Tech**.

Secondary objectives include:

- Applying various Natural Language Processing (NLP) techniques for cleaning and preparing raw text data.
- Implementing the Multinomial Naive Bayes algorithm for text classification.
- Developing an interactive Graphical User Interface (GUI) to demonstrate the model's real-world application.

## 2. Introduction

In the digital age, the volume of online information is overwhelming. Manually sorting and categorizing news articles is inefficient and time-consuming. This project addresses this challenge by building an intelligent system that automates the process of news classification. By applying Natural Language Processing (NLP) techniques and a classic machine learning algorithm, the system can swiftly and accurately assign a category to a given news article based on its content. This has practical applications in news aggregation platforms, content recommendation engines, and digital media monitoring.

## 3. System Requirements

- **Programming Language:** Python 3.x
- **Core Libraries:**
  - **Pandas:** For data loading and manipulation.
  - **Scikit-learn:** For implementing the machine learning model (TF-IDF, Naive Bayes) and evaluation.
  - **NLTK (Natural Language Toolkit):** For text preprocessing tasks like stopword removal.
  - **Streamlit:** For creating the interactive web GUI.
- **Development Environment:** Environment with Python installed.

## 4. Materials and Methods

### 4.1 Dataset

The project uses the **AG News classification dataset**, a standard collection of over 120,000 news articles, each pre-labeled into one of the four target categories.

### 4.2 Text Preprocessing

To prepare the data for the model, the raw text was cleaned. This process involved converting text to lowercase, removing all punctuation and common English stopwords (e.g., "the", "a", "is"), and applying stemming to reduce words to their root form (e.g., "running" to "run").

### 4.3 Feature Extraction: TF-IDF

The cleaned text was converted into a numerical format using the **TF-IDF (Term Frequency-Inverse Document Frequency)** vectorizer. This method calculates a numerical score for each word that reflects its importance to an article, allowing the model to process the text data mathematically.

### 4.4 Classification Model: Multinomial Naive Bayes

The **Multinomial Naive Bayes (MNB)** algorithm was chosen for classification. MNB is a probabilistic classifier based on Bayes' Theorem that is highly efficient and particularly effective for text-based tasks involving word counts.

## 5. Procedure / Implementation Workflow

1. **Environment Setup:** A virtual environment was created to manage project dependencies. All required libraries like pandas, scikit-learn, and streamlit were installed using pip.
2. **Data Loading and Cleaning:** The train.csv file was loaded into a pandas DataFrame. The preprocessing steps mentioned above were applied to the combined title and description of each article.
3. **Model Training:** The preprocessed dataset was split into features (X) and labels (y). The TF-IDF vectorizer was fitted on the text data to create the feature matrix. This matrix was then used to train the Multinomial Naive Bayes model.
4. **Model Saving:** After training, the model and the fitted TF-IDF vectorizer were saved to disk as .pkl files using the pickle library. This allows the trained objects to be loaded later for inference without retraining.
5. **GUI Development:** A user-friendly web application was built using Streamlit. The app.py script loads the saved model and vectorizer, provides a text area for user input, and displays the predicted news category upon clicking a "Classify" button.

## 6. Results and Evaluation

The model's performance was evaluated on a held-out test set (20% of the data).

- **Accuracy:** The model achieved an overall accuracy of approximately **90%**, meaning it correctly classified 9 out of 10 news articles.
- **Classification Report:** A detailed report showed high precision and recall scores (generally between 0.88 and 0.92) for all four classes, indicating that the model is well-balanced and does not have a significant bias towards any single category.
- **GUI:** The final Streamlit application successfully demonstrated the model's functionality, providing instantaneous and accurate predictions for user-provided text.


```
Confusion Matrix:
[[5169  566   72  193]
 [ 473 5197   78  252]
 [  40   42 5820   98]
 [ 294  164  179 5363]]

--- Testing with new headlines ---
Headline: 'NASA launches new mission to explore distant galaxies and stars'
Predicted Category: Sci/Tech

Headline: 'Stock market surges as tech companies report record profits'
Predicted Category: Business

Headline: 'Team wins championship in a thrilling last-minute goal'
Predicted Category: Sports
```



```
@tejashree-dumasia →/workspaces/codespaces-blank (main) $ git push origin main
@tejashree-dumasia →/workspaces/codespaces-blank (main) $ python news_classifier.py
Dataset Head:
                                                text class_name
0  Wall St. Bears Claw Back Into the Black (Reute...   Business
1  Carlyle Looks Toward Commercial Aerospace (Reu...   Business
2  Oil and Economy Cloud Stocks' Outlook (Reuters...   Business
3  Iraq Halts Oil Exports from Main Southern Pipe...   Business
4  Oil prices soar to all-time record, posing new...   Business

Class Distribution:
class_name
Business    30000
Sci/Tech    30000
Sports      30000
World       30000
Name: count, dtype: int64

Preprocessing text data...
Preprocessing complete.

Training the Multinomial Naive Bayes model...
Model training complete.

Model Accuracy: 0.8979

Classification Report:
              precision    recall  f1-score   support

       World       0.86      0.86      0.86      6000
      Sports       0.87      0.87      0.87      6000
    Business       0.95      0.97      0.96      6000
    Sci/Tech       0.91      0.89      0.90      6000

    accuracy                           0.90     24000
   macro avg       0.90      0.90      0.90     24000
weighted avg       0.90      0.90      0.90     24000
```